7. Which of the following could qualify as a top-down control on a grassland community?
   (A) limitation of plant biomass by rainfall amount
   (B) influence of temperature on competition among plants
   (C) influence of soil nutrients on the abundance of grasses versus wildflowers
   (D) effect of grazing by bison on plant species diversity

8. The most plausible hypothesis to explain why species richness is higher in tropical than in temperate regions is that
   (A) tropical communities are younger.
   (B) tropical regions generally have more available water and higher levels of solar radiation.
   (C) higher temperatures cause more rapid speciation.
   (D) diversity increases as evapotranspiration decreases.

9. Community 1 contains 100 individuals distributed among four species: 5A, 5B, 85C, and 5D. Community 2 contains 100 individuals distributed among three species: 30A, 40B, and 30C. Calculate the Shannon diversity index (H) for each community. Which community is more diverse?

Levels 5-6: Evaluating/Creating

10. **DRAW IT** In the Chesapeake Bay estuary, the blue crab is an omnivore that eats eelgrass and other primary producers as well as clams. It is also a cannibal. In turn, the crabs are eaten by humans and by the endangered Kemp's Ridley sea turtle. Based on this information, draw a food web that includes the blue crab. Assuming that top-down control occurs in this system, describe what would happen to the abundance of eelgrass if humans stopped eating blue crabs.

11. **EVOLUTION CONNECTION** Explain why adaptations of particular organisms to competition may not necessarily represent instances of character displacement. What would a researcher have to demonstrate about two competing species to make a convincing case for character displacement?

12. **SCIENTIFIC INQUIRY** An ecologist studying desert plants performed the following experiment. She staked out two identical plots, containing sagebrush plants and small annual wildflowers. She found the same five wildflower species in roughly equal numbers on both plots. She then enclosed one plot with a fence to keep out kangaroo rats, the most common grain-eaters of the area. After two years, four of the wildflower species were no longer present in the fenced plot, but one species had become much more abundant. The control plot had not changed in species diversity. Using the principles of community ecology, propose a hypothesis to explain her results. What additional evidence would support your hypothesis?

13. **WRITE ABOUT A THEME: INTERACTIONS** In Batesian mimicry, a palatable species gains protection by mimicking an unpalatable one. Imagine that individuals of a palatable, brightly colored fly species are blown to three remote islands. The first island has no predators of that species; the second has predators but no similarly colored, unpalatable species; and the third has both predators and a similarly colored, unpalatable species. In a short essay (100–150 words), predict what might happen to the coloration of the palatable species on each island through time if coloration is a genetically controlled trait. Explain your predictions.

14. **SYNTHESIZE YOUR KNOWLEDGE**

Describe two types of ecological interactions that appear to be occurring between the three species shown in this photo. (Look closely to see the three species: a flower, a bee, and a spider!) What morphological adaptation can be seen in the species that is at the highest trophic level in this scene?

For selected answers, see Appendix A.

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